

How is AI bad the environment?

Posted by u/Famous-Channel3027 • 0 points • 14 comments

I keep seeing that ChatGPT and other forms of AI are TERRIBLE for the environment, but every time I ask someone why, they can't give me an answer other than "it uses up so much electricity". Google hasn't helped either....Don't ALL computers use tons of electricity? Isn't AI just software? What am I missing here?

Comments

u/Honest-Bridge-7278 • 1 points • 2025-08-18 11:33 UTC

I think you don't understand computers as much as you think you do, and you need to listen when people answer your question.

u/Famous-Channel3027 • 1 points • 2025-08-18 18:16 UTC

I'm listening! I just ask questions to better understand and I explain my way of thinking to be better understood. People always think I am arguing, but I am not trying to. I have no knowledge of computers and don't claim to. I just see things a certain way and don't understand until I have all of my questions answered.

u/otacon7000 • 14 points • 2025-08-18 05:35 UTC

Three main issues.

- AI needs _a lot_ of computational power, currently this is mostly provided in the form of GPUs; all this hardware needs to be built, requiring resources
- Lots of compute means a lot of electricity usage
- Lots of electricity usage means lots of heat, requiring lots of cooling, which for big data centers usually means a lot of water usage

u/Famous-Channel3027 • -7 points • 2025-08-18 05:52 UTC

More so than the computers and software running our infrastructure, electrical grid, military bases and literally everything else we rely on? It just doesn't add up to me....

u/DiogenesKuon • 1 points • 2025-08-18 17:17 UTC

The problem is that ChatGPT uses about 10x as much energy to answer a query as doing something like a database lookup or a google search. Larger models, and more iterative designs (CoT, Research Mode), may push this significantly higher. This is not a lot of energy for each individual use, but in aggregate it adds up, and if we start replacing a lot of existing tools with AI based tools (which is the trend) we are going to use a lot more energy.

The total amount of energy that datacenter used in the US increased from 60 TWh per year in 2014 to 76 TWh in 2018, a solid but not alarming 26.7% increase in energy usage. But since then we've rapidly increased to 176 TWh in 2023, a much more substantial 131.6% increase. Or to put it another way, in 2018 datacenter were taking up 1.9% of all power in the US, which increased to 4.4% by 2023, and is estimated to continue to grow up to between 6.7% to 12% of all US power. AI is what's driving almost all of this increase in power usage.

This increase in electricity usage will also cause energy prices to spike. Between 2008 and 2021 energy costs have been fairly flat, and have gone from \$0.132 / kWh to \$0.144 / kWh, an increase of 9% over 13 years. Since then it's spiked to \$0.189 / kWh, an increase of 31% in 4 years.

u/pingwing • 7 points • 2025-08-18 06:56 UTC

Yes, they are using very high end chips that use a ton of processing power to run AI. It isn't just "running" AI though, the huge cost is the improving and improving on AI. In order to be "smart" it needs tons of data, there isn't enough data on the planet, so now they have AI create the data that it learns on, creating more processing power.

Ask AI.

u/SadSickSoul • 13 points • 2025-08-18 06:19 UTC

Remember, this is additive - we're doing all that stuff, and now AI. Now that I'm at home I'm looking at sources for things, and there's an interesting article by technology review .com titled "We did the math on AI's energy footprint. Here's the story you haven't heard." that I would highly suggest taking a look through. Here's one paragraph of this extremely comprehensive and, seemingly, pretty even-handed take with caveats throughout on what's known and what's unknown:

"In analyzing both public and proprietary data about data centers as a whole, as well as the specific needs of AI, the researchers came to a clear conclusion. Data centers in the US used somewhere around 200 terawatt-hours of electricity in 2024, roughly what it takes to power Thailand for a year. AI-specific servers in these data centers are estimated to have used between 53 and 76 terawatt-hours of electricity. On the high end, this is enough to power more than 7.2 million US homes for a year."

And that's **now**. They're trying to build more datacenters, put AI into more products and services and generally just try to get even more out, so the energy demands and environmental effects are only going to get more and more crazy.

u/Famous-Channel3027 • 1 points • 2025-08-18 06:24 UTC

Hmmm food for thought. Thank you 😊

u/Felicia_Svilling • -4 points • 2025-08-18 06:14 UTC

Honestly a lot of the environmental complaints are exaggerated because people just don't like genai and want to have more objective reasons for their dislike.

u/Honest-Bridge-7278 • 2 points • 2025-08-18 11:36 UTC

Honestly a lot of environmental complaints are dismissed because people don't understand the extent to which the climate is already

damaged, and they are dickheads.

Genai is bad for more than just the environmental reasons - it is essentially industrialised copyright theft.

u/SadSickSoul • 6 points • 2025-08-18 05:14 UTC

Yes, all computers use electricity. There IS a massive amount of environmental damage that our pre-AI digital infrastructure took, it's true. The problem is that the push towards shoving AI into everything and making it so immediately available all the time is that this one type of application almost overnight exploded the need for electricity and also water to cool the equipment, as far as I understand.

I don't have statistics on how quickly or how much it jumped up, but if we were having trouble managing the environmental impact before, the push for AI increased it on an unimaginable scale, all for tools that have incredibly low value by themselves - once you realize the scale of damage that's being done for the things we use for, it's extremely infuriating. People would rather cook the world for chatbots, AI art or whatever than do the work themselves or hire the people who can.

u/Famous-Channel3027 • 1 points • 2025-08-18 05:51 UTC

So, I use ChatGPT now instead of google because it's not filled with ads and spammy links. It's a million times easier and faster than "googling it". Especially because you don't have reword your question 20 times to actually pull up the proper links. I have fact checked it several times and it's always accurate. Plus, I am a professional cook and AI has been a GODSEND for recipe conversions. I just don't understand how anyone can say it's not useful. It's my absolute favorite technological advancement so far. Sorry, idk why I am defending myself😂 Thank you! One more thing though, please don't think I am an idiot.....how does electricity hurt the environment?

u/SadSickSoul • 6 points • 2025-08-18 05:58 UTC

Well, remember that we're not just turning up a dial and more electricity comes out of a tap of already stored energy. We need to generate electricity such as burning fossil fuels or coal, which incurs its own environmental impact, and then we use up water to generate the steam we use to generate that electricity. So there's a heavy resource cost that vastly increases when AI is so heavily used.

u/Famous-Channel3027 • 1 points • 2025-08-18 06:10 UTC

That makes more sense